

In-Class

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** You may keep the exam sheet for your records.

Part A. Do any five (5) of the following problems:

1. (12) Let $z = 2 - 2i$ and $w = -3\sqrt{3} + 3i$. Write in rectangular form, $a + bi$, each of the following:

a. $\frac{w^2}{z^2}$ b. $(z - \bar{w})(\bar{z} - w)$

2. (12) Prove the following proposition: Let $z, w \in \mathbb{C}$.

Then: $|z + w| \leq |z| + |w|$.

3. (12) Prove the following proposition: Let A be a subset of a metric space (X, d) .

Then: A is open if and only if $A = \text{int } A$

4. (12) Prove the following proposition: Let K be a compact subset of a metric space (X, d) .

Then: If F is closed and $F \subset K$, then F is compact.

5. (12) Prove the following proposition: Let A be a subset of a metric space (X, d) .

Then: For any $x \in X$, $d(x, A) = d(x, \bar{A})$.

6. (12) Prove the following proposition: Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be Lipschitz.

Then: $f + g$ is Lipschitz.

Part B. Do each of the following problems:

7. (10) Provide a counterexample to each of the following assertions:

- a. In a metric space (X, d) , for any $A \subset X$ we have $\text{diam}(\text{int}(A)) = \text{diam}(A)$
- b. In a metric space (X, d) , for any $A \subset X$ we have $\overline{\text{int } A} \subset \text{int } \overline{A}$
- c. For $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$, the function $d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2|$ defines a metric on $\mathbb{R}^2$.
- d. Let (X, d) be a compact metric space and let $f : X \rightarrow \mathbb{R}$ be continuous on X . Then, f is Lipschitz.
- e. Let (X, d) be a metric space and let $f : X \rightarrow \mathbb{R}$ be continuous on X . If A is a open subset of X , then $f(A)$ is a open subset of $\mathbb{R}$.

8. (30) Classification Problem. Correctly identify whether the following subsets of $\mathbb{C}$ are: (a) open; (b) closed; (c) connected; (d) polygonally path connected; (e) compact; (f) complete; (g) bounded; (h) region. You do not need to provide a rationale for your classification. Fill out the classification information on the attached page.

- A. $\{z = x + iy : x > 0, |y| < \frac{1}{|x|}\}$
- B. $\{B(3, 2) \cup B(-3, 2)\} \cap \overline{B(0, 2)}$
- C. $\overline{B(0, 1)} \setminus T$ where T is the interior of the equilateral triangle centered at 0 with vertices $\left\{1, \text{cis} \frac{2\pi}{3}, \text{cis} \frac{4\pi}{3}\right\}$
- D. $B(0, 4) \setminus S$ where $S = \{z : |\text{Re } z| < 2\}$
- E. $\bigcup_{n=1}^{\infty} B_n$, where each $B_n = \overline{B(c_n, \frac{1}{n^2})}$ and the centers c_n satisfy $c_1 = 1, c_2 = c_1 + 1 + \frac{1}{2^2}, c_3 = c_2 + \frac{1}{2^2} + \frac{1}{3^3}, \dots, c_{n+1} = c_n + \frac{1}{n^2} + \frac{1}{(n+1)^2}$ for $n \geq 1$

Classification Table for Problem 8

	open	closed	connected	polygonally path connected	compact	complete	bounded	region
A								
B								
C								
D								
E								